

Module 2

Regular Expressions (Regex)

Process &

Regex is a pattern used to search for and manipulate text.

What can it do?

- ① Find patterns :
- ② Validate text
- ③ Extract text
- ④ Edit text
- ⑤ Generate reports

Regex are patterns that define a series of characters to match, can use quantifiers, pattern collections, special characters. are supported in many programming languages. perl, python, Php, Java, JS etc

Real-time Use :-

- ① Extracting information from web scrapping
- ② Text pre processing for NLP.
- ③ Spam filtering for emails.
- ④ Log analysis: Identify specific events or errors.
- ⑤ Twitter Analysis, #tags

Basics in Regex

- ① Brackets [] :- They are used to specify a disjunction of characters. Using the brackets to put g or G allows us to

Return Capital G or Small g

Ex:- `/[gG] eluvaraj /` → Geluvaraj or geluvaraj

`/[abc] /` → 'a', 'b' or 'c'

`/[1234567890] /` → any Digit

2) Dash (-) :- putting this symbol between any Letters.
allows to return those matches

$/[A-Z]/ \rightarrow$ matches an Upper Case

$/[a-z]/ \rightarrow$ matches an Lower Case

$/[0-9]/ \rightarrow$ matches a ~~single~~ digit 0 to 9.

③ Caret (^) :- Can be used for negation

$/[^A-Z]/ \rightarrow$ not an Upper Case letter

$/[^Ss]/ \rightarrow$ neither 'S' nor 's'

$/[e^]/ \rightarrow$ either 'e' or '^'

④ Question Marks (?) : It marks optionality of the
Previous Expression for instance putting a ? returns the
below

Ex:- $/Geluvaraja?/$

$\rightarrow$ Geluvaraj or Geluvaraja

$/Geluvaraja?a/$

$\rightarrow$ Geluvaraja or Geluvaraja

$/Colou?r/$

$\rightarrow$ Colox or Colours

⑤ Period (.): used to specify any characters between Expressions

Ex:- /beg.n/

Ex:- begin (or) began

Quantifiers

[] ?	occurs 0 or 1 times	<i>abc? → ab followed by zero or one c</i>
[] +	occurs 1 or more times	<i>abc+ → ab followed by one or more c</i>
[] *	occurs 0 or more times	<i>a(bc)* matches followed by zero or more copies of the sequence</i>
[] {n}	occurs n times	<i>abc{2}: ab followed by 2 c</i>
[] {n,}	occurs n or more times	<i>ab{2,} → ab followed by 2 or more c</i>
[] {y, z}	occurs at least y times but less than z times.	<i>abc{2,5} matches ab followed by 2 upto 5 c</i>

Metacharacters: Shortform

\d [0-9]

\D [^ 0-9]

\w [a-z A-Z _ 0-9]

\W [^\w]

\s matches whitespace, tabs and line breaks

Escape Sequence (or)

\ (backword slash).

tells Computer to following Character as search character for '+' '.' '?' etc

Anchors: They are used to assert something about the string (or) matching process.

Ex:- Consider 'A' and '\$'

[^]The : Matches any string that starts with 'The'

end \$: matches any string that ends with 'end'

^ The end \$: Exact string match (starts and ends with 'The end')

Numericals:

① Mobile number $\rightarrow$ start with 8 or 9
and total digit = 10

$[89] [0-9] \{9\}$ $\rightarrow$ 9 times repeat.

$\downarrow$ $\downarrow$

Chosen any 1 digit any combination of all the numbers 0 to 9

② First character Upper case, and it should contain lower case alphabets, only one digit allowed in between.

between

$[A-Z]$ $[a-z]$ * $[0-9]$ $[a-z]$ *

Lower case

No's

Lower case.

③ Email ID: Ex: geluvaraj999@gmail.com
geluvaraj.b@gmail.com

$$[a-z A-Z 0-9 _ - | \cdot] + [@] [a-z] + [\cdot] [a-z] \{ 2, 3 \}$$

So using escape sequence

com
-2] {2, 3}
↓
at least 2 charact
not more than 3